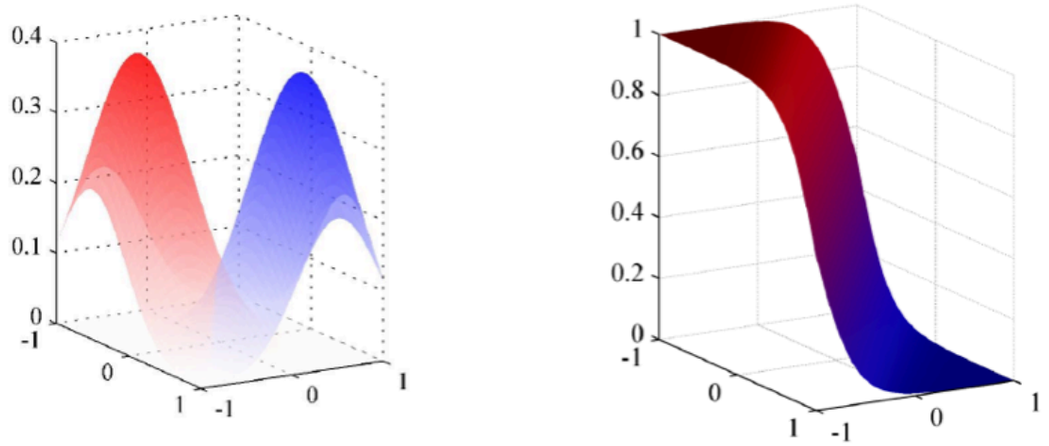
Mixture of Gaussians for Classification

Generative Probabilistic Models

Hoda Eldardiry
Machine Learning

Mixture of Gaussians for Classification

- Generative Probabilistic Models



- Build a classifier by modeling how each class generates data

Lecture Objectives

- Generative vs Discriminative Models
- Bayes Decision Rule
- Gaussian Class-Conditionals
- Logistic sigmoid
- Softmax
- Maximum likelihood learning

Assumptions → Model → Inference → Learning

Generative vs Discriminative Models

- **Generative Model** $P(x, C) = P(C)P(x|C)$
  - Class → Generate x
  - Learns: Prior & Class-conditional distribution
  - ***Generative models explain how data is produced***
  - Examples: Gaussian classifiers, Naive Bayes, LDA, QDA
- **Discriminative Model** $P(C|x)$
  - x → Predict Class
  - ***Discriminative models directly learn decision boundaries***
  - Examples: Logistic Regression, Neural Networks, SVM

Generative Models

- **Probabilistic Generative Models**
  - Can be used for regression & classification
  - Can be used to simulate the creation of a dataset
  - Can generate data similar to training data

- Why do we need to generate data?

Data Generation Applications

- **Text generation**
  — Conversation agent
  — Goal: generate a response similar to responses we might find in a dataset of (message, response) pairs

- **Image generation**
  — Diagnosis from medical images
  — Goal: generate synthetic images to
  (1) Preserve patient privacy
  (2) Learn models to understand cases not in training data

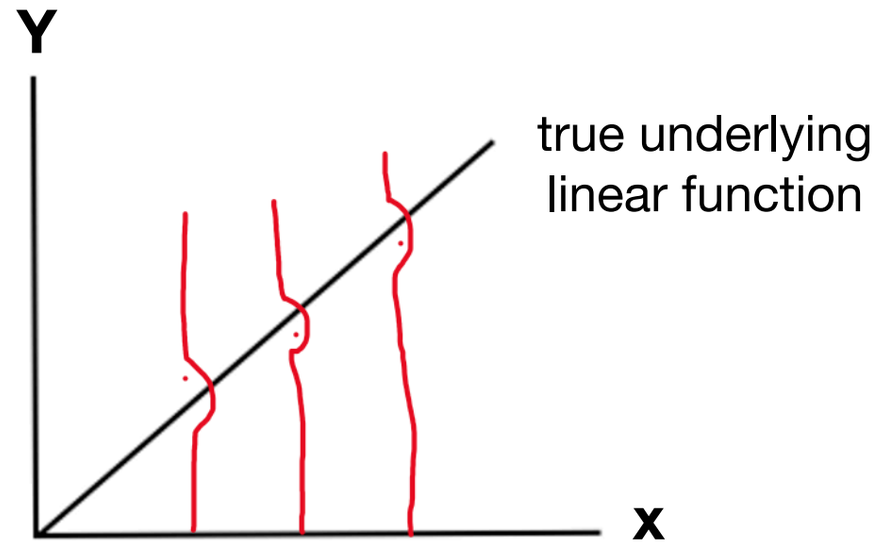
Regression vs Classification

- **Generative Regression Assumptions:**

$$y = f(x) + \epsilon$$

$$\epsilon \sim \mathcal{N}(0, \sigma^2)$$

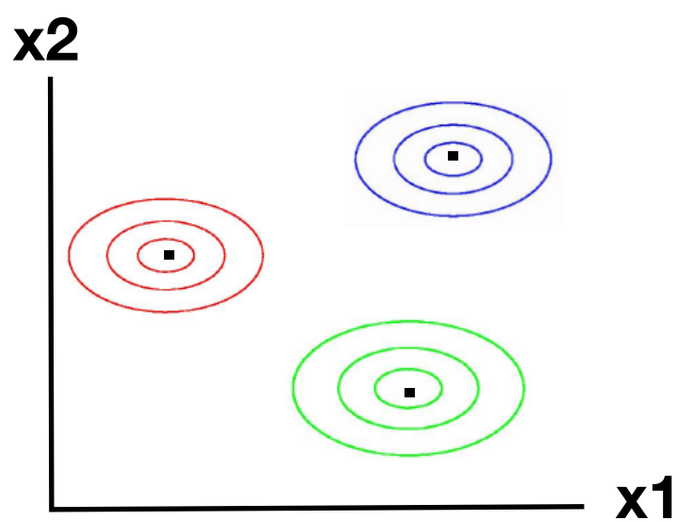
  - Output sampled from Gaussian around a function



- **Generative Classification Assumptions:**

  - Each class generates data from a Gaussian

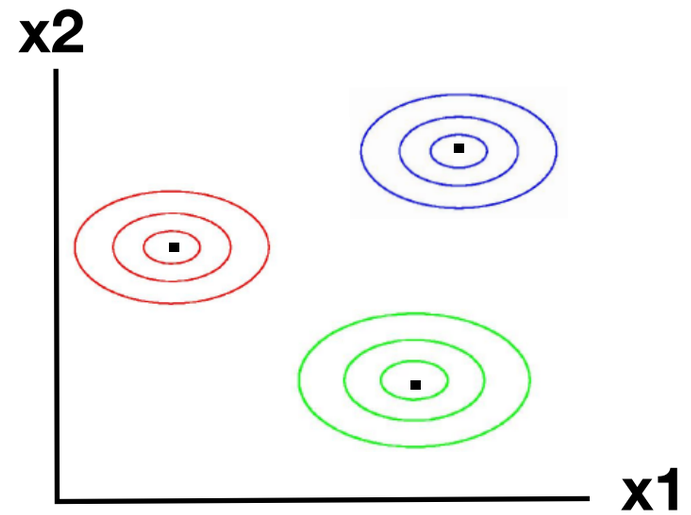
  - Data = noisy measurements around class centers



Bayesian Classification

Classification Setup

- Assume:
  - K classes
  - Input: $\boldsymbol{x} \in \mathbb{R}^d$
  - Output: categorical label C_k
- Goal: Compute $P(C_k|\boldsymbol{x})$



The Three Components

- **Learning: estimates prior & likelihood**
  - **Prior** $P(C_k) = \pi_k$
    - Multinomial distribution $\sum_k \pi_k = 1$
  - **Likelihood (class conditional)** $P(x|C_k)$
    - Assume Gaussian $\mathcal{N}(x|\mu_k, \Sigma_k)$

“can be used to generate points from class C”

- **Inference: computes posterior using Bayes rule**
  - **Posterior** $P(C_k|x) = \frac{P(C_k)P(x|C_k)}{\sum_j P(C_j)P(x|C_j)}$

Bayesian Inference

$$P(C_k|x) = \frac{P(C_k)P(x|C_k)}{\sum_j P(C_j)P(x|C_j)}$$ ⟶ normalizing constant

- Posterior ∝ Prior × Likelihood
- Key idea:
  - *Classification = compare weighted likelihoods*

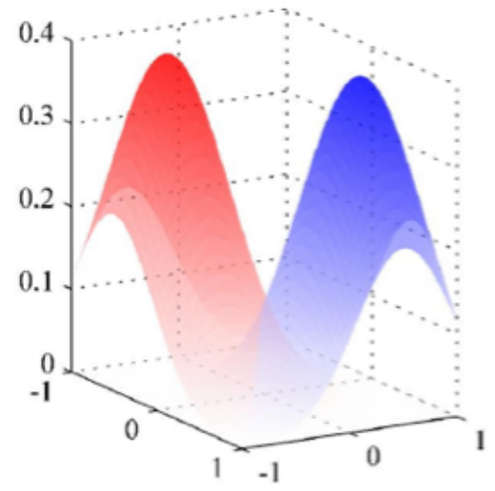
Gaussian Class Conditionals

Gaussian Assumption

- Class conditional distribution: Gaussian

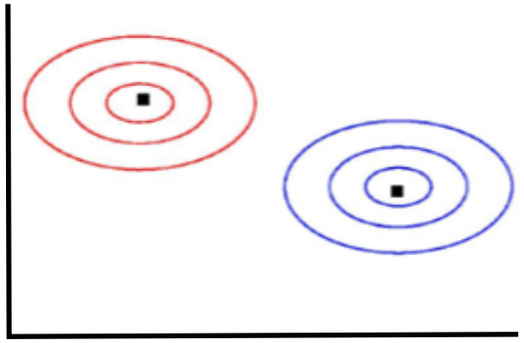
$$P(x|C_k) = \mathcal{N}(x|\mu_k, \Sigma_k)$$

  - Mean = class center
  - Covariance = shape & orientation



- Can be written in exponential form:

$$\Pr(\boldsymbol{x}|c_k) \propto e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_k})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_k})}$$

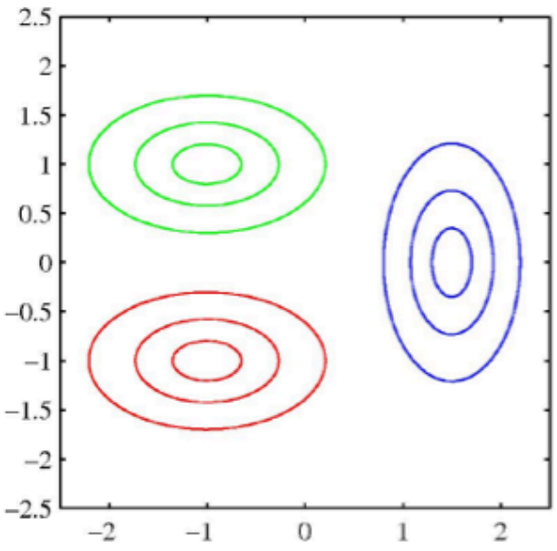


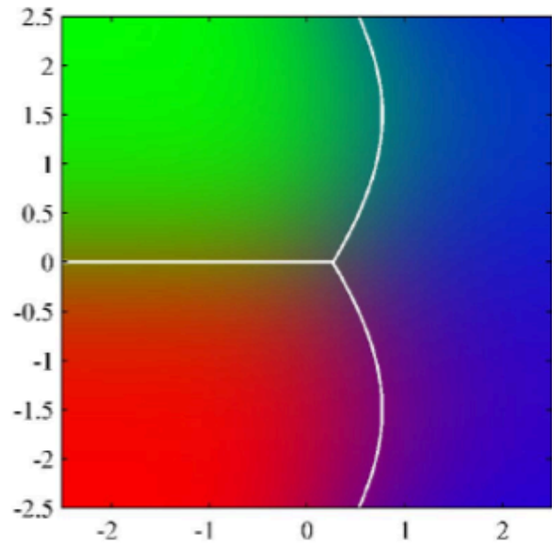
- Note: equal covariance assumption simplifies the exponent

$$\Sigma_1 = \Sigma_2$$

What Covariance Controls

- Spread
- Orientation
- Shape (circular vs elliptical)
- Covariance determines geometry of decision boundary





Binary Class Case → Logistic Sigmoid

Binary Posterior

$$P(C_k|x) = \frac{P(C_k)P(x|C_k)}{\sum_j P(C_j)P(x|C_j)}$$

$$P(C_k) = \pi_k \qquad P(x|C_k) = \mathcal{N}(x|\mu_k, \Sigma)$$

- For 2 classes: $P(C_1|x) = \frac{\pi_1 \mathcal{N}_1}{\pi_1 \mathcal{N}_1 + \pi_2 \mathcal{N}_2}$

$$\mathcal{N}(x|\mu_k, \Sigma_k) = \Pr(\boldsymbol{x}|c_k) \propto e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_k})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_k})}$$

$$\Pr(c_k|\boldsymbol{x}) = \frac{\pi_k e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_k})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_k})}}{\sum_k \pi_k e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_k})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_k})}}$$

Binary Posterior

$$\Pr(c_k|\boldsymbol{x}) = \frac{\pi_k e^{-\frac{1}{2}(x-\mu_k)^T \Sigma^{-1}(x-\mu_k)}}{\sum_k \pi_k e^{-\frac{1}{2}(x-\mu_k)^T \Sigma^{-1}(x-\mu_k)}}$$

$$= \frac{\pi_k e^{-\frac{1}{2}\left(\cancel{x^T\Sigma^{-1}x} - 2\mu_k^T\Sigma^{-1}x + \mu_k^T\Sigma^{-1}\mu_k\right)}}{\sum_k \pi_k e^{-\frac{1}{2}\left(\cancel{x^T\Sigma^{-1}x} - 2\mu_k^T\Sigma^{-1}x + \mu_k^T\Sigma^{-1}u_k\right)}}$$

Consider two classes c_k and c_j

- rewrite exponent: now has 3 terms
- cross 1st term: doesn't depend on class

$$= \frac{1}{1 + \frac{\pi_j e^{\mu_j^T\Sigma^{-1}x - \frac{1}{2}\mu_j^T\Sigma^{-1}\mu_j}}{\pi_k e^{\mu_k^T\Sigma^{-1}x - \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k}}}$$

- simplify other 2 terms in exponent
- divide numerator & denominator by numerator

$$= \frac{1}{1 + e^{-\left(\mu_k^T - \mu_j^T\right)\Sigma^{-1}x + \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k - \frac{1}{2}\mu_j^T\Sigma^{-1}\mu_j - \ln\frac{\pi_k}{\pi_j}}}$$

- Take log-odds and rearrange

$$= \frac{1}{1+e^{-(w^T x + w_0)}}$$

where $\boldsymbol{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_k - \boldsymbol{\mu}_j)$

and $w_0 = -\frac{1}{2}\boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \frac{1}{2}\boldsymbol{\mu}_j^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_j + \ln\frac{\pi_k}{\pi_j}$

Posterior ⇒ Logistic Sigmoid

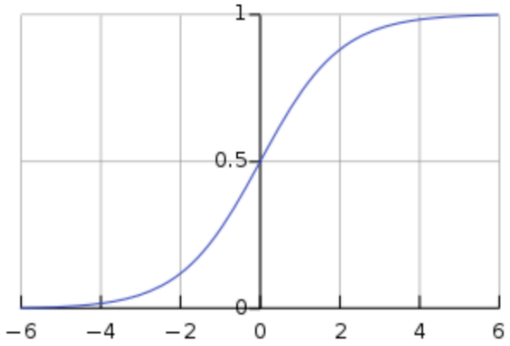
- Derived to:

$$P(C_1|x) = \sigma(w^T x + w_0)$$

- ⇒ Logistic Sigmoid

$$\sigma(a) = \frac{1}{1 + e^{-a}}$$

$$a = w^T x + w_0$$

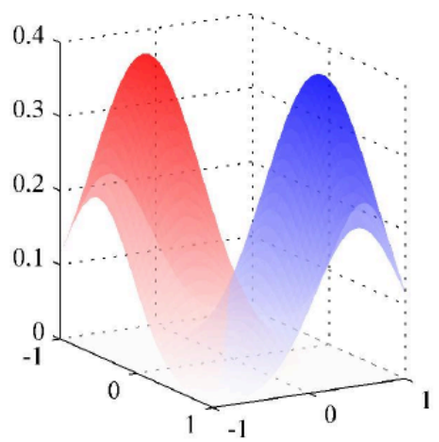


where $\boldsymbol{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_k - \boldsymbol{\mu}_j)$

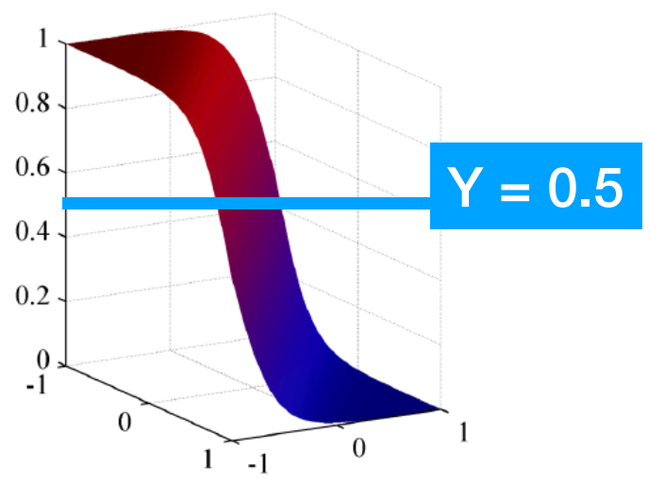
and $w_0 = -\frac{1}{2}\boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \frac{1}{2}\boldsymbol{\mu}_j^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_j + \ln\frac{\pi_k}{\pi_j}$

- *Gaussian generative model ⇒ Logistic posterior*

class conditionals

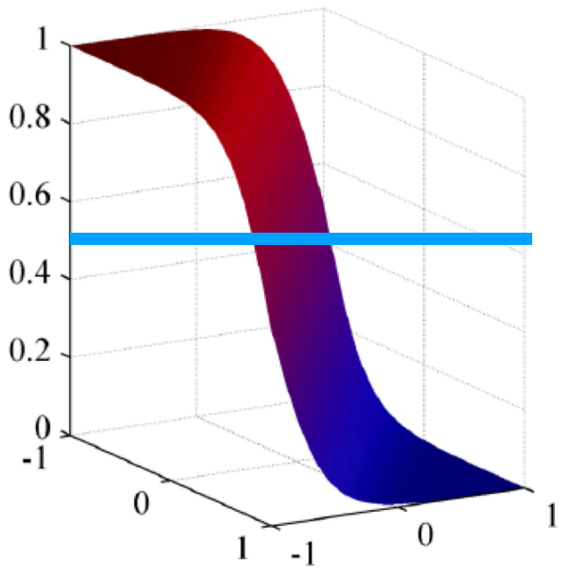


posterior



Binary Prediction

- **Posterior sigmoid:** probability distribution over classes
- Prediction: class of highest probability
- Binary prediction: class of probability >= 0.5



Y = 0.5

$$best\ class = argmax_k \Pr(c_k|\boldsymbol{x})$$

$$= \begin{cases} c_1 & \sigma(\boldsymbol{w}^T\boldsymbol{x} + w_0) \geq 0.5 \\ c_2 & \text{otherwise} \end{cases}$$

Red

Blue

Class boundary: $\sigma\left(\boldsymbol{w}_k^T\overline{\boldsymbol{x}}\right) = 0.5$

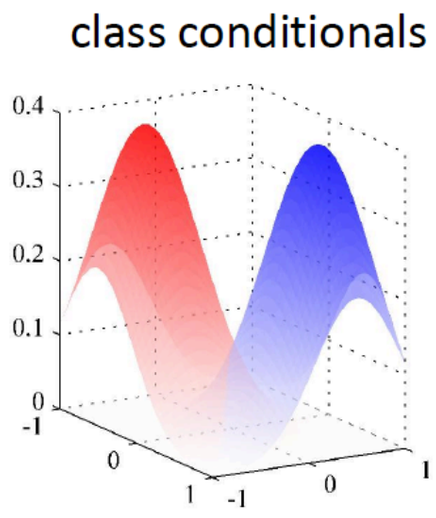
$$\Longrightarrow \frac{1}{1+e^{-\left(\boldsymbol{w}_k^T\overline{\boldsymbol{x}}\right)}} = 0.5$$

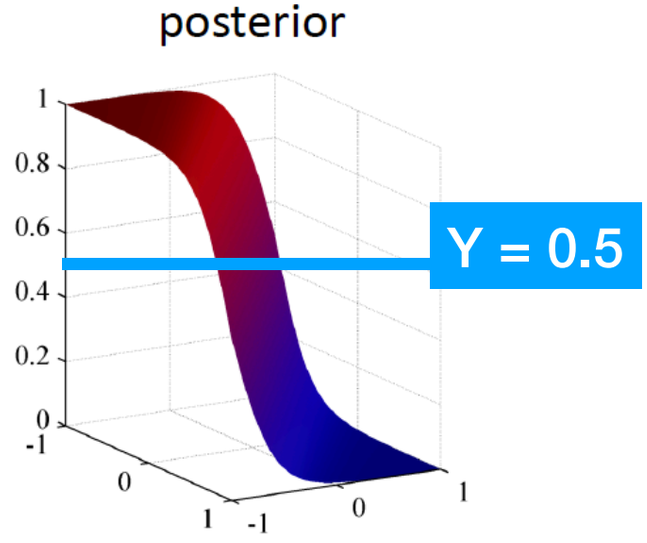
$$\Longrightarrow \boldsymbol{w}_k^T\overline{\boldsymbol{x}} = 0$$

Sigmoid input = 0

$\therefore$ **linear separator** Simple model

Geometry of Decision Boundary





- If covariances are equal:
- → Linear decision boundary $w^T x + w_0 = 0$
- This is Linear Discriminant Analysis (LDA)

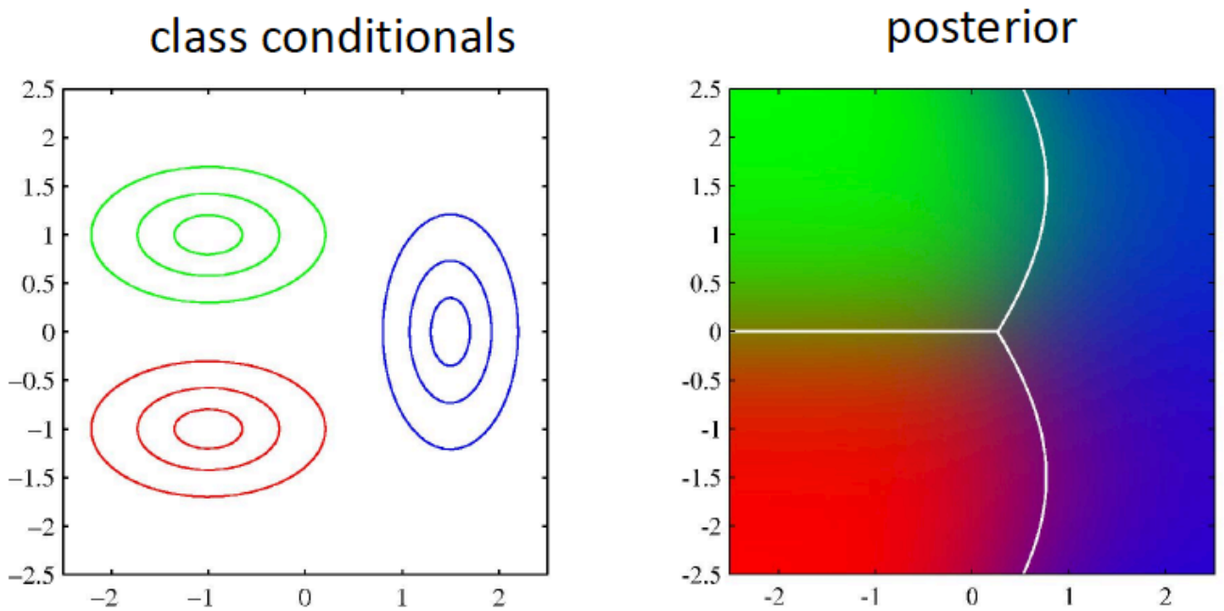
What If Covariances Differ?

- If: $\Sigma_1 \neq \Sigma_2$

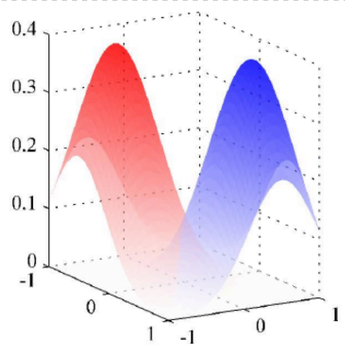
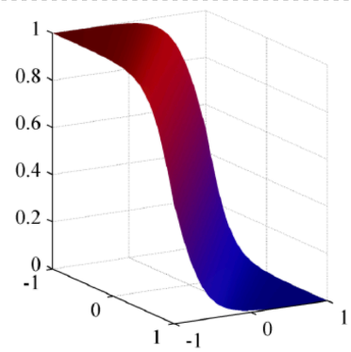
- Then:

  - Decision boundary becomes quadratic

  - This is Quadratic Discriminant Analysis (QDA)



Summary

Model Components	1) Prior P(Ci) "Learning"	2) Likelihood P(x|C) "Learning"	3) Posterior P(C|x) "Inference"
Definition	Prior distribution assumed; without prior knowledge of data	Class conditional distribution of x Given class, likelihood to observe x?	Posterior probability of class C given an input x
Computation			Bayes posterior = prior * likelihood $\Pr(C\|\boldsymbol{x}) = \frac{\Pr(\boldsymbol{x}\|C)\Pr(C)}{\sum_C \Pr(\boldsymbol{x}\|C)\Pr(C)}$
Assumptions & Derivation	Prior: Multinomial distribution $\Pr(C = c_k) = \pi_k$ Classes: finite, categorical	Likelihood: Gaussian distribution $\Pr(\boldsymbol{x}\|c_k) \propto e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_k})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_k})}$ Data: $\boldsymbol{x} \in \Re^d$ Covariance matrix $\boldsymbol{\Sigma}$: same for c1, c2	$\Pr(c_k\|\boldsymbol{x}) = \frac{\pi_k e^{-\frac{1}{2}(x-\mu_k)^T\Sigma^{-1}(x-\mu_k)}}{\sum_k \pi_k e^{-\frac{1}{2}(x-\mu_k)^T\Sigma^{-1}(x-\mu_k)}}$ $= \frac{\pi_k e^{-\frac{1}{2}(\cancel{x^T\Sigma^{-1}x}-2\mu_k^T\Sigma^{-1}x+\mu_k^T\Sigma^{-1}\mu_k)}}{\sum_k \pi_k e^{-\frac{1}{2}(\cancel{x^T\Sigma^{-1}x}-2\mu_k^T\Sigma^{-1}x+\mu_k^T\Sigma^{-1}u_k)}}$ Consider two classes c_k and c_j $= \frac{1}{1+\frac{\pi_j e^{\mu_j^T\Sigma^{-1}x-\frac{1}{2}\mu_j^T\Sigma^{-1}\mu_j}}{\pi_k e^{\mu_k^T\Sigma^{-1}x-\frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k}}}$ $= \frac{1}{1+e^{-(\mu_k^T-\mu_j^T)\Sigma^{-1}x+\frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k-\frac{1}{2}\mu_j^T\Sigma^{-1}\mu_j-\ln\frac{\pi_k}{\pi_j}}}$
Inference: Binary Classification			Posterior: Logistic Sigmoid $\Pr(c_k\|\boldsymbol{x}) = \frac{1}{1+e^{-(\boldsymbol{w}^T\boldsymbol{x}+w_0)}}$ $\boldsymbol{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_k - \boldsymbol{\mu}_j)$ $w_0 = -\frac{1}{2}\boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \frac{1}{2}\boldsymbol{\mu}_j^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_j + \ln\frac{\pi_k}{\pi_j}$
Plot			

Multi-Class Case → Softmax

Posterior for K Classes

- Consider Gaussian conditional distributions with identical Σ — Covariance matrix

$$\Pr(c_k|\boldsymbol{x}) = \frac{\Pr(c_k)\Pr(\boldsymbol{x}|c_k)}{\sum_j \Pr(c_j)\Pr(\boldsymbol{x}|c_j)}$$

$$= \frac{\pi_k e^{-\frac{1}{2}(x-\mu_k)^T \boldsymbol{\Sigma}^{-1}(x-\mu_k)}}{\sum_j \pi_j e^{-\frac{1}{2}(x-\mu_j)^T \boldsymbol{\Sigma}^{-1}(x-\mu_j)}}$$

Simplify D & N
N: linear expression in x
D: linear expression in x & summation over all classes j

$$= \frac{\pi_k e^{-\frac{1}{2}\left(-2\mu_k^T\boldsymbol{\Sigma}^{-1}x+\mu_k^T\boldsymbol{\Sigma}^{-1}\mu_k\right)}}{\sum_j \pi_j e^{-\frac{1}{2}\left(-2\mu_j^T\boldsymbol{\Sigma}^{-1}x+\mu_j^T\boldsymbol{\Sigma}^{-1}\mu_j\right)}}$$

$$= \frac{e^{\mu_k^T\boldsymbol{\Sigma}^{-1}x-\frac{1}{2}\mu_k^T\boldsymbol{\Sigma}^{-1}u_k+\ln\pi_k}}{\sum_j e^{\mu_j^T\boldsymbol{\Sigma}^{-1}x-\frac{1}{2}\mu_j^T\boldsymbol{\Sigma}^{-1}\mu_j+\ln\pi_j}} = \frac{e^{w_k^T\bar{x}}}{\sum_j e^{w_j^T\bar{x}}} \Longrightarrow \text{softmax}$$

where $\boldsymbol{w}_k^T = (-\frac{1}{2}\boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \ln\pi_k\,,\ \boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1})$

W_0

Other entries in vector w:
Coefficients of x

Gaussian generative model ⇒ Softmax classifier

Why Softmax?

- Softmax = smooth version of argmax
- Gives probabilities
- Differentiable
- Used in neural networks

Multi-class Prediction

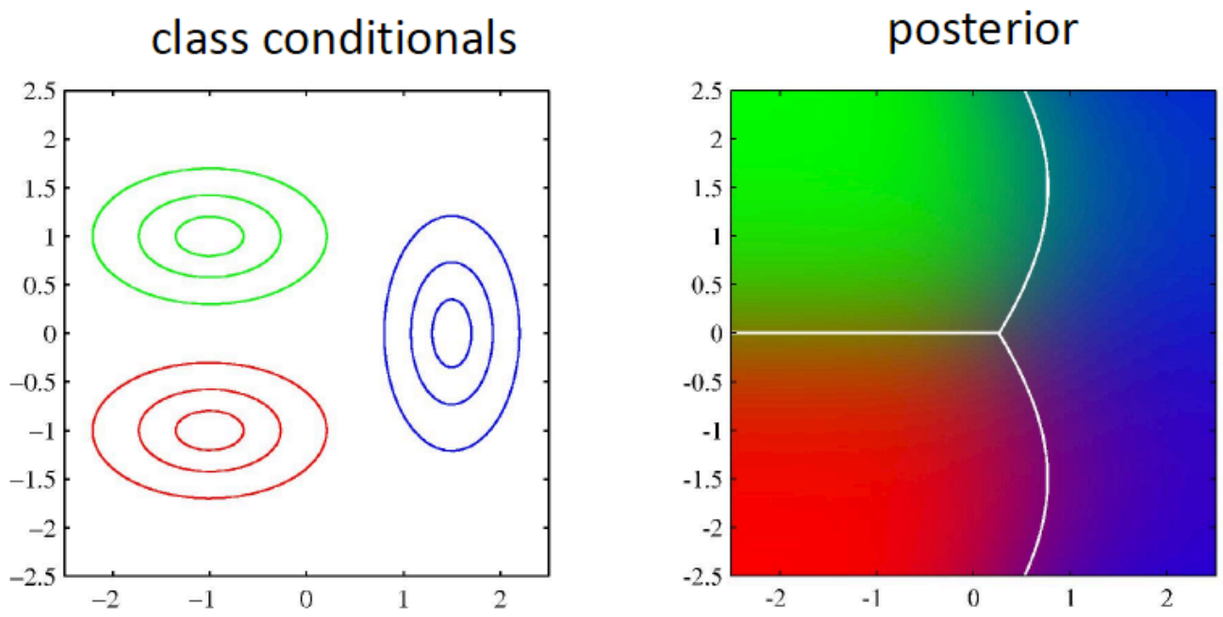
- When there are several classes, the posterior is a **softmax** (generalization of the sigmoid)
- Softmax distribution: $\Pr(c_k|\boldsymbol{x}) = \dfrac{e^{f_k(\boldsymbol{x})}}{\sum_j e^{f_j(\boldsymbol{x})}}$
- Argmax distribution:

$$\Pr(c_k|\boldsymbol{x}) = \begin{cases} 1 & \text{if } k = \underline{argmax}_j\, f_j(x) \\ 0 & \text{otherwise} \end{cases}$$

$$= \lim_{base\to\infty} \frac{base^{f_k(x)}}{\sum_j base^{f_j(x)}}$$

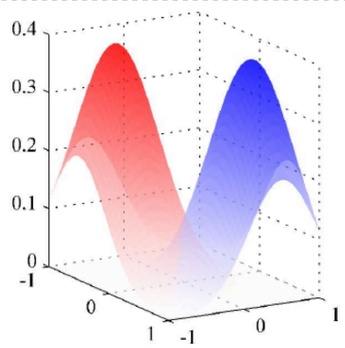
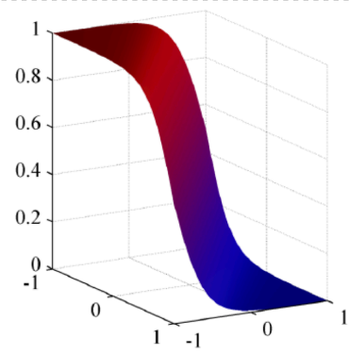
$$\approx \frac{e^{f_k(x)}}{\sum_j e^{f_j(x)}} \quad \text{(softmax approximation)}$$

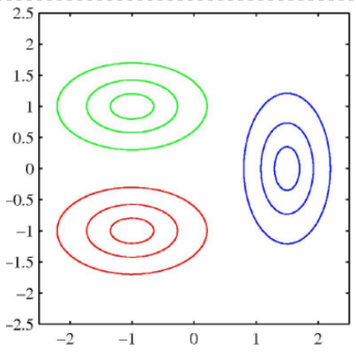
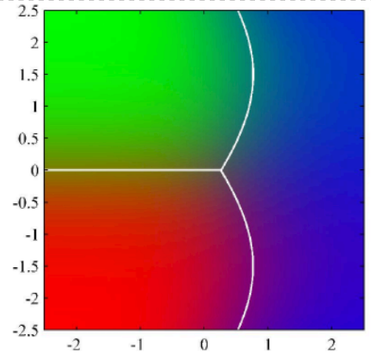
Geometry of Decision Boundary

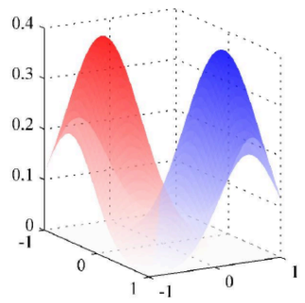
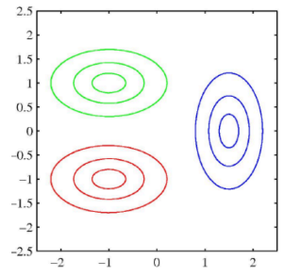
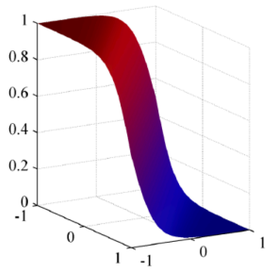
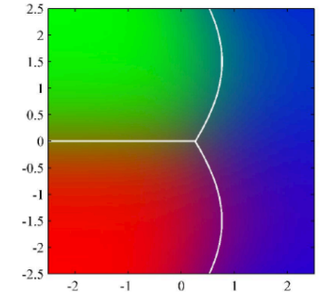


- Covariance matrix determines shape of gaussian

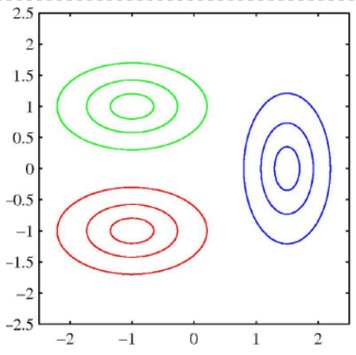
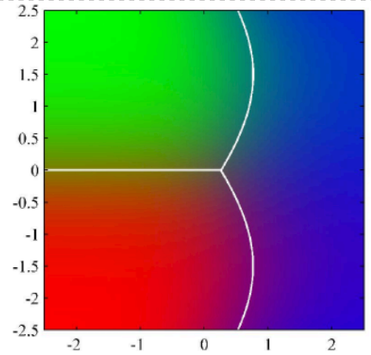
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Definition	Prior distribution assumed; without prior knowledge of data	Class conditional distribution of x given class, likelihood to observe x?	Posterior probability of class C given an input x
Computation			Bayes: posterior=prior x likelihood $\Pr(C\|\boldsymbol{x}) = \frac{\Pr(\boldsymbol{x}\|C)\Pr(C)}{\sum_C \Pr(\boldsymbol{x}\|C)\Pr(C)}$
Assumptions & Derivation	prior: multinomial distribution $\Pr(C = c_k) = \pi_k$ classes: finite, categorical	likelihood: Gaussian distribution $\Pr(\boldsymbol{x}\|c_k) \propto e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_k})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_k})}$ data: $\boldsymbol{x} \in \Re^d$ covariance matrix $\boldsymbol{\Sigma}$ same for c1, c2	$\Pr(c_k\|\boldsymbol{x}) = \frac{\pi_k e^{-\frac{1}{2}(x-\mu_k)^T\Sigma^{-1}(x-\mu_k)}}{\sum_k \pi_k e^{-\frac{1}{2}(x-\mu_k)^T\Sigma^{-1}(x-\mu_k)}}$ $= \frac{\pi_k e^{-\frac{1}{2}(\cancel{x^T\Sigma^{-1}x}-2\mu_k^T\Sigma^{-1}x+\mu_k^T\Sigma^{-1}\mu_k)}}{\sum_k \pi_k e^{-\frac{1}{2}(\cancel{x^T\Sigma^{-1}x}-2\mu_k^T\Sigma^{-1}x+\mu_k^T\Sigma^{-1}u_k)}}$ Consider two classes c_k and c_j $= \frac{1}{1+\frac{\pi_j e^{\mu_j^T\Sigma^{-1}x-\frac{1}{2}\mu_j^T\Sigma^{-1}\mu_j}}{\pi_k e^{\mu_k^T\Sigma^{-1}x-\frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k}}}$ $= \frac{1}{1+e^{-(\mu_k^T-\mu_j^T)\Sigma^{-1}x+\frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k-\frac{1}{2}\mu_j^T\Sigma^{-1}\mu_j-\ln\frac{\pi_k}{\pi_j}}}$
Inference: Binary Classification			Posterior: Logistic Sigmoid $\Pr(c_k\|\boldsymbol{x}) = \frac{1}{1+e^{-(\boldsymbol{w}^T\boldsymbol{x}+w_0)}}$ $\boldsymbol{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_k - \boldsymbol{\mu}_j)$ $w_0 = -\frac{1}{2}\boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \frac{1}{2}\boldsymbol{\mu}_j^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_j + \ln\frac{\pi_k}{\pi_j}$
Plot			

Model Components	1) Prior P(Ci) "Learning"	2) Likelihood P(x|C) "Learning"	3) Posterior P(C|x) "Inference"
Definition	Prior distribution assumed; without prior knowledge of data	Class conditional distribution of x given class, likelihood to observe x?	Posterior probability of class C given an input x
Computation			Bayes: posterior=prior x likelihood $\Pr(C\|\boldsymbol{x}) = \frac{\Pr(\boldsymbol{x}\|C)\Pr(C)}{\sum_C \Pr(\boldsymbol{x}\|C)\Pr(C)}$
Assumptions & Derivation	prior: multinomial distribution $\Pr(C = c_k) = \pi_k$ classes: finite, categorical	likelihood: Gaussian distribution $\Pr(\boldsymbol{x}\|c_k) \propto e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_k})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_k})}$ data: $\boldsymbol{x} \in \Re^d$ covariance matrix $\boldsymbol{\Sigma}$: similar c1, c2, different for c3	$= \frac{\pi_k e^{-\frac{1}{2}(x-\mu_k)^T\Sigma^{-1}(x-\mu_k)}}{\sum_j \pi_j e^{-\frac{1}{2}(x-\mu_j)^T\Sigma^{-1}(x-\mu_j)}}$ $= \frac{\pi_k e^{-\frac{1}{2}(-2\mu_k^T\Sigma^{-1}x+\mu_k^T\Sigma^{-1}\mu_k)}}{\sum_j \pi_j e^{-\frac{1}{2}(-2\mu_j^T\Sigma^{-1}x+\mu_j^T\Sigma^{-1}\mu_j)}}$ $= \frac{e^{\mu_k^T\Sigma^{-1}x-\frac{1}{2}\mu_k^T\Sigma^{-1}u_k+\ln\pi_k}}{\sum_j e^{\mu_j^T\Sigma^{-1}x-\frac{1}{2}\mu_j^T\Sigma^{-1}\mu_j+\ln\pi_j}}$
Inference: Multi-class Classification			Posterior: Softmax $\Pr(c_k\|\boldsymbol{x}) = \frac{e^{\boldsymbol{w}_k^T\bar{\boldsymbol{x}}}}{\sum_j e^{\boldsymbol{w}_j^T\bar{\boldsymbol{x}}}}$ $\boldsymbol{w}_k^T = (-\frac{1}{2}\boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \ln\pi_k,\ \boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1})$
Plot			

Model Components	Binary Classification	Multi-class Classification
Class Conditional Distribution	2 similar conditional distribution covariance matrix 	2 similar conditional distribution covariance matrix 1 different conditional distribution covariance matrix 
Classifier (separator) form	Linear separator between classes of similar covariance matrix 	Linear separator between classes of similar covariance matrix Quadratic separator between classes of different covariance matrix 
Posterior Distribution	Logistic Sigmoid $\Pr(c_k\|\boldsymbol{x}) = \frac{1}{1+e^{-(\boldsymbol{w}^T\boldsymbol{x}+w_0)}}$ $\boldsymbol{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_k - \boldsymbol{\mu}_j)$ $w_0 = -\frac{1}{2}\boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \frac{1}{2}\boldsymbol{\mu}_j^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_j + \ln\frac{\pi_k}{\pi_j}$	Softmax Distribution $\Pr(c_k\|\boldsymbol{x}) = \frac{e^{f_k(\boldsymbol{x})}}{\sum_j e^{f_j(\boldsymbol{x})}}$
Argmax Distribution	$\Pr(c_k\|\boldsymbol{x}) = \begin{cases} c_1 & \sigma(\boldsymbol{w}^T\boldsymbol{x}+w_0) \geq 0.5 \\ c_2 & \text{otherwise} \end{cases}$	$\Pr(c_k\|\boldsymbol{x}) = \begin{cases} 1 & \text{if } k = argmax_j\, f_j(x) \\ 0 & \text{otherwise} \end{cases}$ $= \lim_{base\to\infty} \frac{base^{f_k(x)}}{\sum_j base^{f_j(x)}}$ $\approx \frac{e^{f_k(x)}}{\sum_j e^{f_j(x)}}$

Learning (Parameter Estimation) using Maximum Likelihood

Model Components	1) Prior P(Ci) "Learning"	2) Likelihood P(x|C) "Learning"	3) Posterior P(C|x) "Inference"
Definition	Prior distribution assumed; without prior knowledge of data	Class conditional distribution of x given class, likelihood to observe x?	Posterior probability of class C given an input x
Computation			Bayes: posterior=prior x likelihood $\Pr(C\|\boldsymbol{x}) = \frac{\Pr(\boldsymbol{x}\|C)\Pr(C)}{\sum_C \Pr(\boldsymbol{x}\|C)\Pr(C)}$
Assumptions & Derivation	prior: multinomial distribution $\Pr(C = c_k) = \pi_k$ classes: finite, categorical	likelihood: Gaussian distribution $\Pr(\boldsymbol{x}\|c_k) \propto e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_k})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_k})}$ data: $\boldsymbol{x} \in \Re^d$ covariance matrix $\boldsymbol{\Sigma}$: similar c1, c2, different for c3	$= \frac{\pi_k e^{-\frac{1}{2}(x-\mu_k)^T\boldsymbol{\Sigma}^{-1}(x-\mu_k)}}{\sum_j \pi_j e^{-\frac{1}{2}(x-\mu_j)^T\boldsymbol{\Sigma}^{-1}(x-\mu_j)}}$ $= \frac{\pi_k e^{-\frac{1}{2}(-2\mu_k^T\boldsymbol{\Sigma}^{-1}x+\mu_k^T\boldsymbol{\Sigma}^{-1}\mu_k)}}{\sum_j \pi_j e^{-\frac{1}{2}(-2\mu_j^T\boldsymbol{\Sigma}^{-1}x+\mu_j^T\boldsymbol{\Sigma}^{-1}\mu_j)}}$ $= \frac{e^{\mu_k^T\boldsymbol{\Sigma}^{-1}x-\frac{1}{2}\mu_k^T\boldsymbol{\Sigma}^{-1}u_k+\ln\pi_k}}{\sum_j e^{\mu_j^T\boldsymbol{\Sigma}^{-1}x-\frac{1}{2}\mu_j^T\boldsymbol{\Sigma}^{-1}\mu_j+\ln\pi_j}}$
Inference: Multi-class Classification			Posterior: Softmax $\Pr(c_k\|\boldsymbol{x}) = \frac{e^{\boldsymbol{w}_k^T\bar{\boldsymbol{x}}}}{\sum_j e^{\boldsymbol{w}_j^T\bar{\boldsymbol{x}}}}$ $\boldsymbol{w}_k^T = (-\frac{1}{2}\boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \ln\pi_k,\ \boldsymbol{\mu}_k^T\boldsymbol{\Sigma}^{-1})$
Plot			

Classification Model: Mixture of Gaussian

- **Inference**
  Compute posterior $\Pr(c_k|\boldsymbol{x})$ for 2 or multiple classes

- **Learning**
  Estimate model parameters for:

  - Prior distribution $\Pr(c_k)$

  - Conditional distribution $\Pr(\boldsymbol{x}|c_k)$

Model Parameters

Learning	Prior Distribution Prior probability for each class	Likelihood/Conditional Distribution
Probability	$\Pr(c_k)$	$\Pr(\boldsymbol{x}\|c_k)$
Distribution **(binary classification)** **Probability for each class**	**Multinomial (Binomial)** $\Pr(c_1) = \pi,$ $\Pr(c_2) = 1 - \pi$	**Gaussian** $\Pr(\boldsymbol{x}\|c_1) \propto e^{-\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)}$ $\Pr(\boldsymbol{x}\|c_2) \propto e^{-\frac{1}{2}(x-\mu_2)^T\Sigma^{-1}(x-\mu_2)}$
Parameters	Probability of class 1 π	Mean μ_1, μ_2 Covariance matrix Σ Assume same

LEARN: Estimate model parameters: GIVEN A DATASET, estimate π μ_1, μ_2 Σ

Learning: Parameter Estimation

- Estimate parameters by
  - **Maximum likelihood**
  - Maximum a posteriori
  - Bayesian learning

- **Maximum likelihood:**
  - Simplest, most commonly used, especially with Mixtures of Gaussians

Maximum Likelihood

- Likelihood: $$L(\theta) = \prod_{n=1}^{N} P(x_n, y_n|\theta)$$

- Take log: $$\log L = \sum_{n=1}^{N} \log P(x_n, y_n|\theta)$$

- Log simplifies product → sum

Maximum Likelihood Solution

Likelihood:

$$\mathrm{L}(\mathbf{X}, \mathbf{y}) = \Pr(\boldsymbol{X}, \boldsymbol{y} | \pi, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}) = \prod_n [\pi N(\boldsymbol{x}_n | \boldsymbol{\mu}_1, \boldsymbol{\Sigma})]^{y_n} [(1-\pi) N(\boldsymbol{x}_n | \boldsymbol{\mu}_2, \boldsymbol{\Sigma})]^{1-y_n}$$

$y_n \in \{0,1\}$

- Problem becomes: find parameters that maximize <u>likelihood of data</u>
- Define Likelihood function for dataset (X, y) given model parameters π $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$ $\boldsymbol{\Sigma}$

- For a given data point x,
  - [1] If x belongs to c1, class conditional is Gaussian1 (mean 1)
  - [2] If x belongs to c2, class conditional is Gaussian2 (mean 2)

- Without knowledge of c, Likelihood should still use correct Gaussian (mean) in each c case
  - Define a single expression that combines [1] & [2]
  - Define Likelihood to use correct Gaussian expression
    - Label classes as 0 and 1
    - Use class as exponent
    - The correct Gaussian term is activated with the corresponding class

- This expression represents the probability of obtaining a point with a certain class
  - Given n data points in the given dataset, the likelihood is the product of probability expression of all data points

Maximum Likelihood Solution

Likelihood:

$$\mathrm{L}(\mathbf{X}, \mathbf{y}) = \Pr(\boldsymbol{X}, \boldsymbol{y} | \pi, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}) = \prod_n [\pi N(\boldsymbol{x}_n | \boldsymbol{\mu}_1, \boldsymbol{\Sigma})]^{y_n} [(1-\pi) N(\boldsymbol{x}_n | \boldsymbol{\mu}_2, \boldsymbol{\Sigma})]^{1-y_n}$$

$y_n \in \{0,1\}$

- Find parameters that maximize this likelihood

=> Optimization: maximize L(**X**, **y**)
=> Maximizing a product (non-linear)
=> Complex optimization
=> Simplify
=> Log (expression)
=> Log does not change the maximum
=> Product becomes a sum

Maximum Likelihood Solution

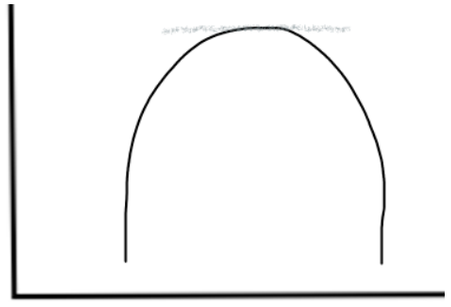
ML hypothesis:

$$< \pi^*, \boldsymbol{\mu}_1^*, \boldsymbol{\mu}_2^*, \boldsymbol{\Sigma}^* > =$$

$$argmax_{\pi,\boldsymbol{\mu}_2,\boldsymbol{\mu}_2,\boldsymbol{\Sigma}} \sum_n y_n \left[\ln \pi - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_1)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_1)\right]$$

$$+(1 - y_n)\left[\ln(1 - \pi) - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_2)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_2)\right]$$

- Optimization is reduced to Log-likelihood maximization
  - Log cancels the exponential
  - Resulting expression has a form that is concave
  - Concave optimization has a single global optimum



- Maximizing log-likelihood corresponds to a curve that has a concave shape
  - Log-likelihood maximization of concave shape
  - Easy optimization function
  - Set derivative to zero & isolate parameters

- **Solve once for each parameter [to estimate each parameter]:**
  - **Take derivative w.r.t. parameter**
  - **Set derivative to zero**
  - **Isolate parameter**

Solve for π

ML hypothesis:

$$< \pi^*, \boldsymbol{\mu}_1^*, \boldsymbol{\mu}_2^*, \boldsymbol{\Sigma}^* > =$$

$$argmax_{\pi,\boldsymbol{\mu}_2,\boldsymbol{\mu}_2,\boldsymbol{\Sigma}} \sum_n y_n \left[\ln \pi - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_1)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_1)\right]$$
$$+(1 - y_n)\left[\ln(1-\pi) - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_2)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_2)\right]$$

- Set derivative to 0

$$0 = \frac{\partial \ln L(\boldsymbol{X},\boldsymbol{y})}{\partial \pi}$$

$$\Rightarrow 0 = \sum_n y_n \left[\frac{1}{\pi}\right] + (1 - y_n)\left[-\frac{1}{1-\pi}\right]$$

$$\Rightarrow 0 = \sum_n y_n (1 - \pi) + (1 - y_n)(-\pi)$$

Isolate π

$$\Rightarrow \sum_n y_n = \pi(\sum_n y_n + \sum_n (1 - y_n))$$

$$\Rightarrow \sum_n y_n = \pi N \quad \text{(where } N \text{ is the \# of training points)}$$

π : probability of class 1

π : fraction of data of class 1

$$\therefore \frac{\sum_n y_n}{N} = \pi$$

? **Sample Mean (Empirical Mean)**

Without derivation, probability of a class corresponds to fraction of data belonging to the class
Which is what maximum likelihood estimator computes
Formal derivation confirms an intuitive approach

Solve for $\boldsymbol{\mu_1, \mu_2}$

ML hypothesis:

$$< \pi^*, \boldsymbol{\mu}_1^*, \boldsymbol{\mu}_2^*, \boldsymbol{\Sigma}^* > =$$

$$argmax_{\pi,\boldsymbol{\mu}_2,\boldsymbol{\mu}_2,\boldsymbol{\Sigma}} \sum_n y_n \left[\ln \pi - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_1)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_1)\right]$$
$$+(1 - y_n)\left[\ln(1-\pi) - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_2)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_2)\right]$$

Y: target label given in training data

$$0 = \partial \ln L(\boldsymbol{X}, \boldsymbol{y}) / \partial \boldsymbol{\mu}_1$$

$$\Longrightarrow 0 = \sum_n y_n[-\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_1)]$$

$$\Longrightarrow \sum_n y_n \boldsymbol{x}_n = \sum_n y_n \boldsymbol{\mu}_1$$

$$\Longrightarrow \sum_n y_n \boldsymbol{x}_n = N_1 \boldsymbol{\mu}_1$$

Isolate $\boldsymbol{\mu}_1$

Yn: Label 1 => class 1

$$\therefore \frac{\sum_n y_n \boldsymbol{x}_n}{N_1} = \boldsymbol{\mu}_1$$

Similarly: $\frac{\sum_n (1-y_n) \boldsymbol{x}_n}{N_2} = \boldsymbol{\mu}_2$

Sample Mean (Empirical Mean)

where N_1 is the # of data points in class 1

N_2 is the # of data points in class 2

This finds best values for both means that jointly maximizes likelihood
In other settings, both means may be in the same term
Then we must solve a system of linear equations

Solve for $\boldsymbol{\Sigma}$

ML hypothesis:

$$< \pi^*, \boldsymbol{\mu}_1^*, \boldsymbol{\mu}_2^*, \boldsymbol{\Sigma}^* > =$$

$$argmax_{\pi,\boldsymbol{\mu}_2,\boldsymbol{\mu}_2,\boldsymbol{\Sigma}} \sum_n y_n \left[\ln \pi - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_1)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_1)\right]$$
$$+(1 - y_n)\left[\ln(1-\pi) - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_2)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_2)\right]$$

$$\frac{\partial \ln L(\boldsymbol{X},\boldsymbol{y})}{\partial \boldsymbol{\Sigma}} = 0$$

$$\Longrightarrow \cdots$$

Linear convex combination of empirical covariance matrices for each class weighted by number of points in each class

$$\Longrightarrow \boxed{\Sigma = \frac{N_1}{N}\boldsymbol{S}_1 + \frac{N_2}{N}\boldsymbol{S}_2}$$

where $\boldsymbol{S}_1 = \frac{1}{N_1}\sum_{n \in c_1}(\boldsymbol{x}_n - \boldsymbol{\mu}_1)(\boldsymbol{x}_n - \boldsymbol{\mu}_1)^T$

$$\boldsymbol{S}_2 = \frac{1}{N_2}\sum_{n \in c_2}(\boldsymbol{x}_n - \boldsymbol{\mu}_2)(\boldsymbol{x}_n - \boldsymbol{\mu}_2)^T$$

($\boldsymbol{S}_k$ is the empirical covariance matrix of class k)

MLE Solutions Summary

- **Estimate Prior** $\hat{\pi}_k = \frac{N_k}{N}$
  - Fraction of data in class k
  - Intuitive and optimal under MLE
- **Estimate Mean** $\hat{\mu}_k = \frac{1}{N_k} \sum_{n:y_n=k} x_n$
  - Class empirical mean
- **Estimate Covariance** $\hat{\Sigma} = \frac{1}{N} \sum_{k} \sum_{n:y_n=k} (x_n - \mu_k)(x_n - \mu_k)^T$
  - Equal covariances
  - Weighted combination of class covariances

Big Picture & Connections

Key Insight

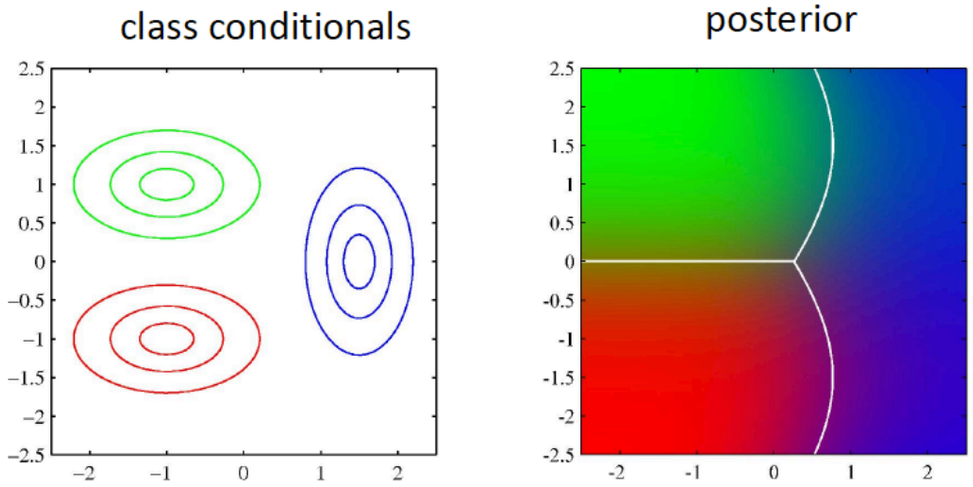
- Generative Gaussian Classifier
  ↓
- Produces
  - Logistic regression (binary)
  - Softmax classifier (multi-class)
  - Under equal covariance assumption
- Gaussian Assumption
  - → Bayes Rule
  - → Linear in x
  - → Sigmoid / Softmax

Generative vs Logistic Regression

- **Generative (Gaussian)**
  - Models P(x)
  - Strong assumptions
  - Works well with small datasets
- **Logistic Regression**
  - Models P(C)
  - Fewer assumptions
  - Better with large datasets

Decision Boundary Summary

- Covariance Assumption



- Same Σ: Linear boundary

- Different Σ: Quadratic boundary

Conceptual Wrap-Up

Model Pipeline

- Learning:
  - Estimate priors
  - Estimate means
  - Estimate covariance
- Inference:
  - Compute posterior
  - Argmax

When is this Model Good?

- Good when:
    - Classes approximately Gaussian
    - Data not extremely high-dimensional
    - Interpretability matters
    - Small dataset

Class Assumption

- This assumes:
  - 1 Gaussian per class
- Other techniques assume:
  - Multiple Gaussians per class